

API BUDDY - API Testing & Management Platform

Project Overview

API Buddy is a comprehensive, web-based API testing and management platform designed for developers, QA engineers, and API developers to test, debug, and manage REST APIs with ease. Built with modern technologies like React, TypeScript, and Vite, this application provides an intuitive interface for making HTTP requests, analyzing responses, and maintaining API collections—all within a single, cohesive dashboard.

The platform combines the functionality of tools like Postman with integrated AI-powered insights, allowing users to test APIs, view execution history, and generate automatic test results. It's a full-stack solution with user authentication, request management, and real-time response visualization.

Key Features & Functionalities

1. API Request Testing

- **Multiple HTTP Methods:** Support for GET, POST, PUT, PATCH, and DELETE requests
- **Dynamic URL Configuration:** Enter custom API endpoints with full URL support
- **Parameter Management:** Add, edit, and manage query parameters with enable/disable toggles
- **Headers Management:** Configure custom headers for API requests
- **Authentication Support:** Bearer token authentication for secured endpoints
- **Request Body:** Send JSON payloads for POST, PUT, and PATCH requests
- **Real-time Response:** Get instant feedback with response status, headers, and body data

2. Response Analysis & Testing

- **Automatic Test Generation:** AI-powered automatic test results for every API response
- **Status Code Validation:** Automatic checking if the response is successful (2xx) or erroneous
- **Performance Metrics:** Track response time and payload size for performance analysis
- **Response Visualization:** View formatted JSON responses with syntax highlighting
- **Test Result Cards:** Display comprehensive test results with explanations and suggestions

3. Request History

- **Complete History Log:** Maintain a detailed history of all API requests made in the session
- **Quick Access:** Reuse previous requests from history for repeated testing
- **Pagination Support:** Navigate through large history with pagination
- **Request Details:** View full request and response details from any historical entry
- **Test Results:** Review test results from previous API calls

4. Collection Management

- **Create Collections:** Organize API endpoints into logical collections

- **Route Management:** Group related API endpoints within collections
- **Save & Organize:** Save frequently used API requests for quick access
- **Collection View:** Browse and manage all your API collections in one place

5. User Authentication & Profiles

- **User Registration:** Create new user accounts with secure authentication
- **Login System:** Secure login with authentication tokens
- **Protected Routes:** Role-based access control for authenticated users
- **User Profile:** Manage user profile information and preferences

6. AI-Powered test case Panel

- **Chat Interface:** Interactive chat interface with AI assistant
- **Request Assistance:** Get AI suggestions for API testing and debugging
- **Test Insights:** Receive AI-powered explanations for test failures
- **Interactive Panel:** Toggle panel for quick assistance

7. Auto-Schema Generator

- **Automatic Schema Detection:** Generate API schemas automatically from requests
- **Schema Visualization:** View and edit API schemas for documentation
- **Schema Management:** Store and organize API schemas

Technical Architecture

Frontend Stack

- **Framework:** React 18+ with TypeScript
- **Build Tool:** Vite (high-performance build tool)
- **UI Components:** shadcn/ui (customizable UI components)
- **Form Handling:** React Hook Form with Zod validation
- **State Management:** Zustand (lightweight state management)
- **Data Fetching:** TanStack React Query (React Query) for server state management
- **HTTP Client:** Axios for API requests
- **Routing:** React Router v6 for client-side navigation
- **Styling:** Tailwind CSS with PostCSS
- **Animations:** Framer Motion for smooth animations
- **Icons:** Lucide React icons
- **Linting:** ESLint with TypeScript support